Kauda Baji: Proposed Rules

For review — July 2026

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Why this document

I'm currently working on a hobby project to build an AI agent that plays Kauda Baji. The idea is to build a complete model of the game in code, then design AI agents that can play it. In the future, time permitting, there may even be a robot that plays the game with you in real life.

I've progressed though making models for the board, player, pieces, and moves. The next step is defining the rules. This must be completed before moving to designing the AI agent.

And this is where I need your help!

The ask is for you to please read this document and provide feedback if you agree or disagree with each rule. Because a computer will be playing using these rules, they must be fully unambiguous.

I would be grateful for your time and feedback.

If you'd like to follow my progress, the code lives at `https://github.com/MGNaik/KaudiPandit`.

Terms used

Our game is played in Gujarati, but code is typically written in English so I've had to come up with some terms for common concepts. Here are the terms I will use in this document, and their meanings.

Piece
: One of 6 movable objects belonging to one player.

Tuple
: A player's own pieces stacked on the same square and moving together as a unit.

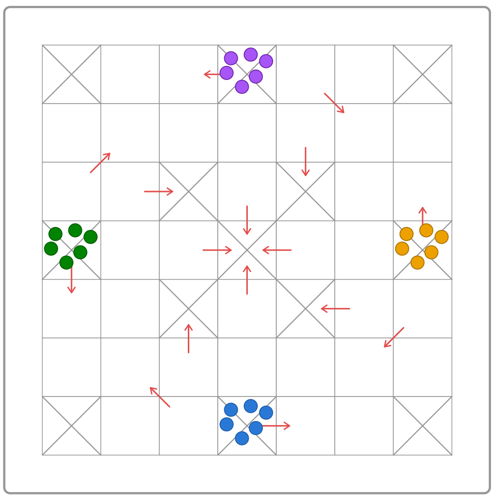
Tuple Size
: The number of pieces in a tuple. A tuple of size n is called an n-tuple — a 1-tuple is a single piece on its own, a 2-tuple has size 2, a 4-tuple has size 4, etc.

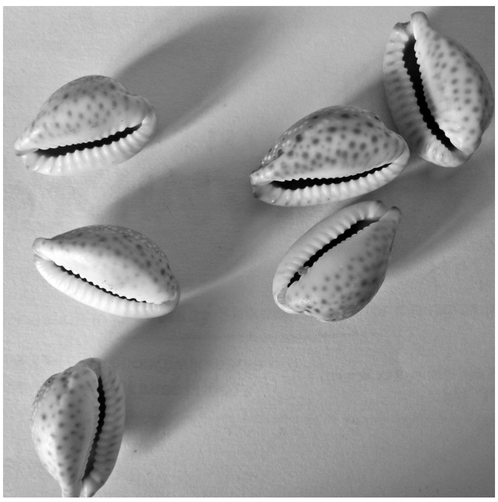
Kaudi
: One of six cowrie shells thrown together to generate a move. Plural: *kaudis*.

Safe square
: A square marked with a corner-to-corner cross. The starting squares and the centre square are safe.

Kill
: Removing an opponent's tuple from the board. Killed pieces go back to their owner's starting square and begin their journey again.

The board and pieces





The board is a 7×7 grid of squares. Four players sit at the four edges; each player owns the edge nearest them.

Moves are generated by throwing a set of six *kaudis* (cowrie shells), and the outcome of the throw is used to move the players' pieces around the board.

Each player has six pieces of their own colour. Every player's pieces start on the same square: the crossed square in the middle of that player's own edge. This is the player's **starting square**. From there, pieces travel around and inward, and all four players finish on the single square at the very centre of the board.

The objective of the game is to get all six pieces from the starting square to the centre square. The first player to do so wins.

The rules

Rule 1 — Determining the first player

Before play begins, every player throws their kaudis once. Whoever rolls the highest value goes first. Play then proceeds **anticlockwise** around the table from there.

Example: Red throws 4, Blue throws 6, Green throws 2, Yellow throws 12. Yellow goes first (12 beats everything), then play continues anticlockwise from Yellow's seat.

Example: Red throws 5, Blue throws 5, Green throws 6, Yellow throws 3. Green goes first with the single highest value, 6.

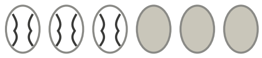
If two or more players tie for the highest value, only those tied players throw again to break the tie; whoever wins that re-throw goes first. Players who weren't tied are not involved in the re-throw.

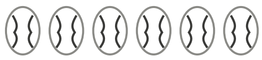
Example: Red throws 6, Blue throws 6, Green throws 4, Yellow throws 2. Red and Blue are tied for highest and throw again, just the two of them, until one beats the other.

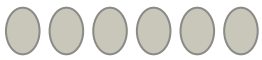
Rule 2 — Throwing the kaudis

A turn begins with a throw of all six kaudis. The value of a throw is the number of kaudis that land mouth-up, except that a throw with **no** kaudis mouth-up is worth **12** rather than 0. So a single throw is worth 1, 2, 3, 4, 5, 6, or 12.

A throw of 6 or 12 earns the player an extra throw, and the value of that extra throw is **banked** alongside the first. If a player throws 6 or 12 three times in a row, they forfeit the rest of their turn — none of the banked values may be used.


3 mouth-up → value 3


6 mouth-up → value 6 (bonus: throw again)


0 mouth-up → value 12 (bonus: throw again)

Example: A player throws 6 (a bonus), throws again and gets 4 — since 4 isn't 6 or 12, the turn stops there with two values banked to spend: 6 and 4. A player who instead throws 6, then 12, then 6 has hit a bonus three times in a row and forfeits the whole turn — none of those three values may be used.

Rule 3 — Using a throw to move

Each value produced during a turn moves one piece that many squares. If a turn produces more than one value, they may all be spent on a single piece (added together), or spread across several different pieces, in any combination the player chooses — but a single value cannot be split between two pieces. A throw of 6 and 3 in the same turn, for example, can move one piece 9 squares in total, or two different pieces 6 and 3 squares each, but the 6 and the 3 can never themselves be divided further.

Example: A player throws 6 (a bonus, so they throw again), then 12 (another bonus, so they throw once more), then 4 — not a 6 or 12, so the turn stops there with three values banked: 6, 12, and 4. Legal ways to spend them include: one piece moves 22 (6+12+4); one piece moves 18 (6+12) and a second piece moves 4; or three different pieces move 6, 12, and 4 respectively. Not legal: splitting the 6 into a 4-and-2 to feed two pieces.

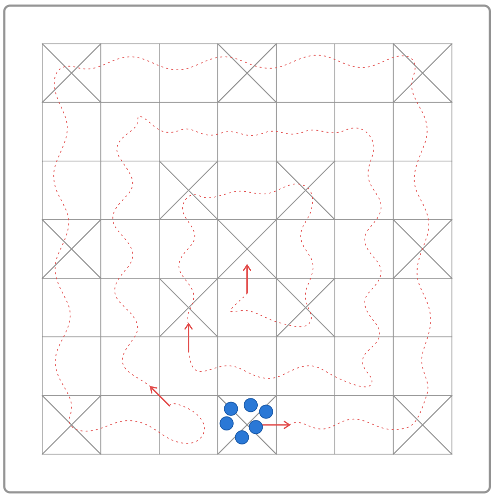
Example: A player throws 5 on their first throw. Since 5 is neither 6 nor 12, no bonus throw follows — the turn produces just that one value. The whole 5 must go to a single piece; there's nothing to spread, since there's only the one value.

Rule 4 — The movement path: anticlockwise out, clockwise in, three arrows to the centre

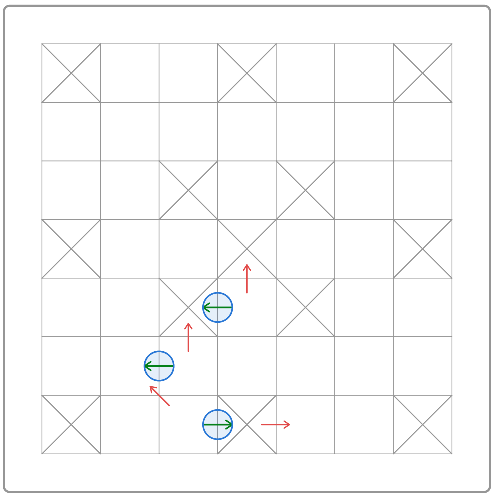
Every piece follows the same route from its starting square to the centre. The route is identical for all four players, just rotated to begin at each player's own edge.

A piece begins on its **starting square** — the crossed square in the middle of the player's edge — and travels anticlockwise around the outer ring of the board. At the **first arrow**, it turns inward onto the first inner band and continues **clockwise** around that band. At the **second arrow**, it turns inward again onto the second inner band, still travelling clockwise. The **third and final arrow** then carries the piece onto the centre square, completing its journey.

At each of the three arrows, a piece is not forced inward: its owner may choose to follow the arrow onto the next band, or keep the piece moving straight ahead on its current band instead, for strategic reasons. (The first arrow carries one further condition — see Rule 5.)



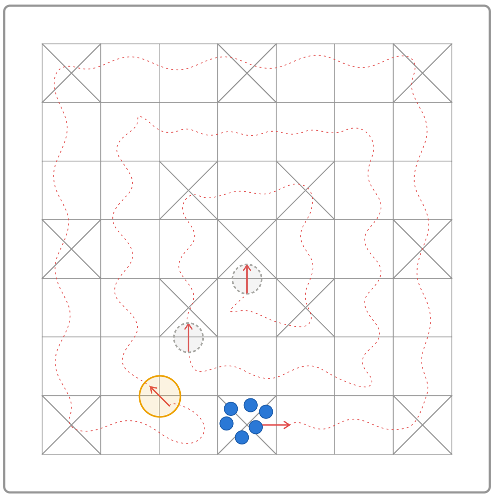
The full route: anticlockwise on the outer ring, clockwise on each inner band, three arrows carrying the piece to the centre.



Turning at each arrow is optional — shown on the bottom player's path; the same three points apply, rotated, for every player.

Rule 5 — The Markut rule: a kill is required to enter the first inner band

A player may not turn any piece inward at the **first** arrow until that player has killed at least one opposing piece, at any point earlier in the game. This is a one-off unlock, not a per-piece cost: once a player has made their first kill, *all* of that player's pieces may enter the first inner band freely from then on. No kill is required to enter the second inner band or the centre — the requirement applies only to the first arrow.



Arrow 1 is gated by this rule (highlighted yellow); arrows 2 and 3 (dashed grey rings) are not.

Rule 6 — A kill earns a bonus throw

Making a kill — landing on and killing an opponent (Rule 11 or 13), or killing on exit (Rule 12) — earns the killer an immediate extra throw. Besides this and throwing 6 or 12 (Rule 2), nothing else grants a bonus throw.

A bonus throw earned from a kill does **not** count towards the limit in Rule 2 of forfeiting after rolling 6 or 12 three times in a row — that limit is only about consecutive 6s and 12s *from the kaudis themselves*. A kill's bonus throw doesn't reset or extend that count either way.

Rule 7 — Tuples form only when a player chooses to move pieces off together

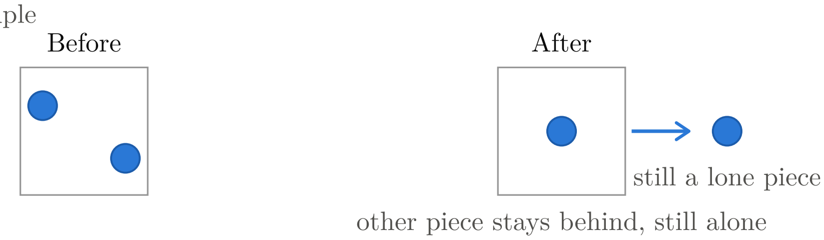
When one of a player's pieces or tuples lands on a square already holding one or more of that **same** player's own pieces or tuples, nothing merges automatically — they simply share the square as separate pieces, exactly as before. A tuple only forms at the moment its owner chooses to move two or more of them off that square **together**, as a single unit; forming and moving happen in the same action, and that move must obey the divide rule (Rule 9) for the combined size. If the pieces are instead moved off individually, or not moved together, no tuple forms.

Once formed, a tuple moves, and is killed, together as one unit (see *Tuple* in Terms used). Tuple size is a plain sum of the pieces merged into it, with no cap other than a player's own six pieces — two dugalu (2-tuples) can combine into a chogalu (4-tuple), and so on up to a single 6-tuple holding all of a player's pieces at once.

A tuple can form on any square other than a player's own **starting square** — ordinary squares and safe squares alike. The starting square is the one exception: pieces waiting there remain single pieces no matter how many share it, and can never be moved off together as a tuple from there. A tuple can only begin forming once a piece has left the starting square.

Moved together: tuple forms

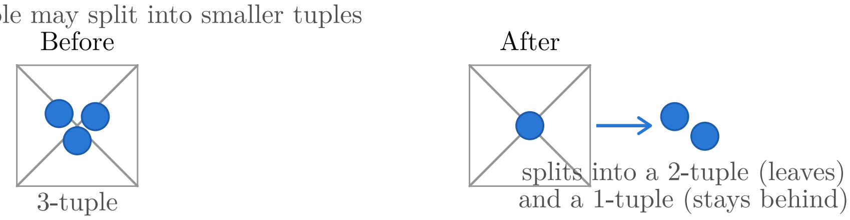
Moved separately: no tuple



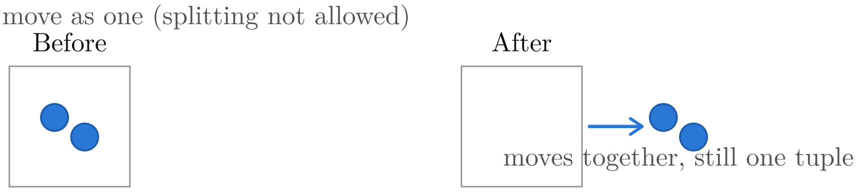
Rule 8 — Tuples may only dissolve on a safe square

A tuple may split, but only while it occupies a safe square. The split need not go all the way down to single pieces — a tuple may divide into any combination of smaller tuples and pieces, with some parts leaving the square and others staying behind. Everywhere else on the board, a tuple must move and be killed as a single unit until it next reaches a safe square.

On a safe square, a tuple may split into smaller tuples



Off a safe square: must move as one (splitting not allowed)



Rule 9 — Moving a tuple: the divide rule

A tuple of size n can only be moved using a throw value that n divides without remainder. A 2-tuple needs a throw of 2, 4, 6, or 12; a 4-tuple needs a throw of 4 or 12; a lone piece (a 1-tuple) can move on any throw, since 1 divides everything. A throw value that doesn't divide evenly by a tuple's size simply cannot be used to move that tuple.

Tuple size	**Usable throw values (out of 1,2,3,4,5,6,12)**
1-tuple (lone piece)	1, 2, 3, 4, 5, 6, 12 — any throw
2-tuple	2, 4, 6, 12
3-tuple	3, 6, 12
4-tuple	4, 12
5-tuple	5 — only an exact throw of 5 works
6-tuple	6, 12

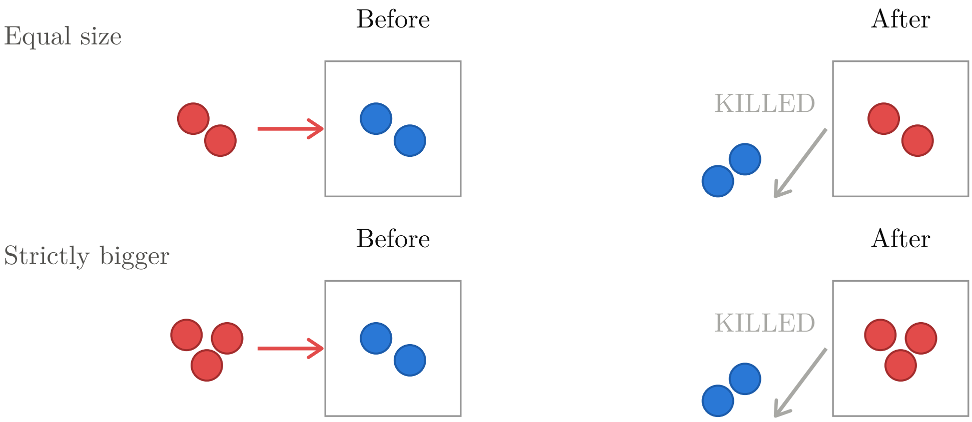
Rule 10 — Safe squares: no kills, ever

Twelve squares carry a corner-to-corner cross, and one more sits at the centre — these are the **safe squares**, marked on the board. Every other square is ordinary.

No kill of any kind can happen on a safe square. Any number of tuples, of any sizes, from any players, can share a safe square. Landing there kills nothing, and leaving a shared safe square kills nothing.

Rule 11 — Landing on an opponent: equal-or-bigger kills, smaller coexists

If your tuple lands on a square holding an opponent's tuple, and your tuple is the **same size or bigger**, the opponent's tuple is killed. Your tuple takes the square; their pieces go back to their start.



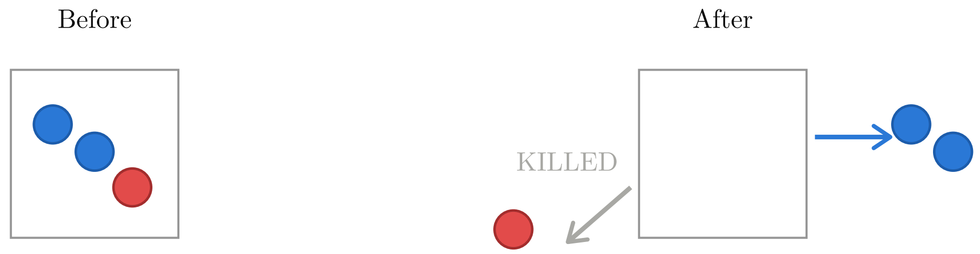
The same happens if the landing tuple is strictly bigger — a Red 3 landing on a Blue 2 kills it just the same.

If instead your tuple lands on a square holding an opponent's tuple and yours is **strictly smaller**, nothing is killed. Both tuples share the square.

Notice what this means: whenever two opponents share a square, the tuple that was there first is always the bigger one.

Rule 12 — Leaving a shared square: the bigger tuple kills on exit

When two opponents share a square and the **strictly bigger** tuple moves away, it kills the smaller tuple as it leaves. (This rule is for exactly two parties sharing a square, each with a single tuple — see Rule 14 if that one opponent has multiple separate tuples there instead, or Rule 15 when a third or later opponent is involved.)



The reverse is safe: if the *smaller* tuple moves away first, no kill happens — it simply escapes.

The smaller tuple sits under threat until either it leaves, or the bigger tuple leaves and kills it on the way out.

Rule 13 — Landing on an opponent with multiple separate tuples: the largest one you can beat dies

An opponent can have more than one of their own tuples sitting on the same square without them being merged (see Rule 7) — for example a 1-tuple and a 3-tuple side by side, never moved together. When your tuple lands on that square, look at all of their tuples that are of **equal or lesser size** than yours — the ones you're big enough to beat — and kill only the **largest** of those. Every other tuple of theirs is safe: any of theirs smaller than the one you killed are spared, and any of theirs bigger than yours were never at risk to begin with.

Lander size 3: kills the 3-tuple (the largest one it can beat)

Lander size 2: kills the 1-tuple instead (3-tuple ineligible)

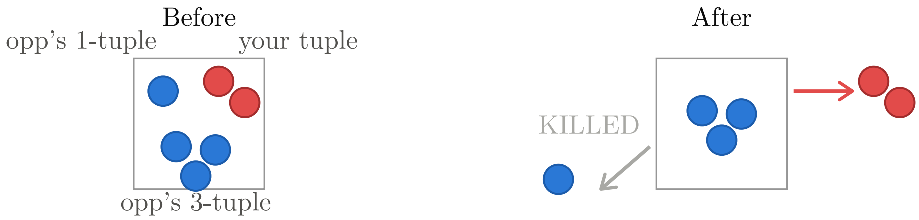
This is still a **two-player** interaction — one opponent owning several tuples, not several different opponents (Rule 15 covers that case). This rule is specifically about **landing** — see Rule 14 for what happens when someone leaves a square like this instead.

Rule 14 — Leaving a square where multiple tuples remain: the same largest-you-can-beat logic

When a tuple leaves a square, it can kill the **largest** of the **opposing** tuples left behind (belonging to a different player) that is **strictly smaller** than the leaving tuple — matching Rule 12, where only a *strictly bigger* tuple kills on exit (a tie never kills). Every opposing tuple that's the same size as the leaver or bigger is safe, and any opposing tuple smaller than the one killed is spared too. This applies no matter **which** tuple leaves — yours, or one of an opponent's own separate tuples — and no matter how many tuples are left behind.

This is a generalisation of Rule 12, not a separate mechanic: with only one tuple left behind, "the largest remaining tuple strictly smaller than the leaver" is just that one tuple if it qualifies, so Rule 12 is the two-tuple special case of this rule.

Your tuple (size 2) leaves: their 3-tuple is safe (bigger), kills their 1-tuple instead



Their 3-tuple leaves instead: kills YOUR tuple (their 1-tuple was never a target)

Rule 15 — Three or more players on one square: no kills at all, landing or leaving

Rules 11–14 are only for a square shared by exactly **one** opponent (whether that opponent has one tuple there or several unmerged ones). Once a square already holds tuples from **two or more different** opponents, everything changes, for **both landing and leaving**: **no kill can happen there, regardless of size**, in either direction.

Landing: a tuple landing on such a square simply joins the pile, no matter how big or small it is compared to what's already there.

Lander bigger than both existing: still no kill

Lander smaller than both: no kill either

Leaving: if a tuple currently shares a square with **two or more** other opponents' tuples and moves away, it kills nobody on exit either. Exit kills (Rule 12) only ever happen when exactly one opponent's tuple is present.

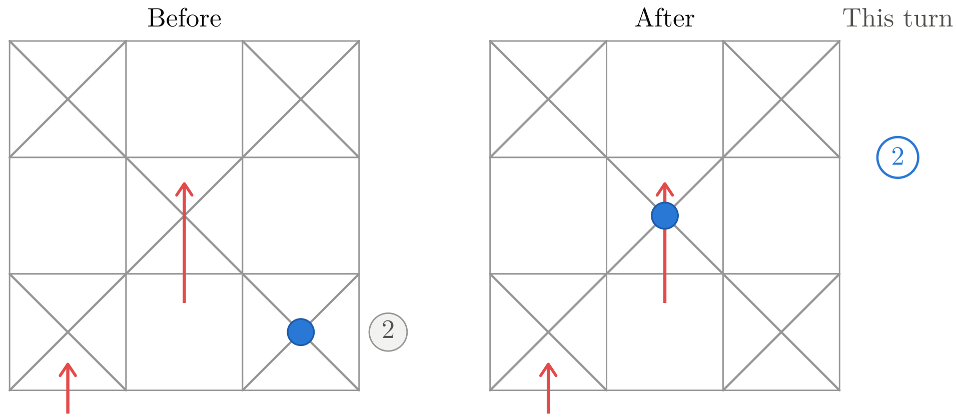
Rule 16 — Reaching the centre: exact throw required

Landing on the centre square requires an **exact** throw — a piece may not overshoot it. “Exact” isn’t limited to a single value either: any combination of this turn’s values that sums to precisely the remaining distance can be used, the same way Rule 3 lets values be summed onto one piece anywhere else on the board. Beyond that, it breaks down into a few cases:

1. **A value smaller than what’s needed** doesn’t reach the centre, but moving the piece part-way with it is still an ordinary, legal move. Like any other legal value, it is not optional: it **must** be used if it can legally move **any** piece on the board — including this one, part-way.
2. **A value larger than what’s needed** cannot land *this* piece on the centre, and is never forced onto it.
3. If that larger value can legally move a **different** one of the player’s pieces, it **must** be used there instead.
4. If no piece on the board — this one included — can legally use it, only then is it forfeited.

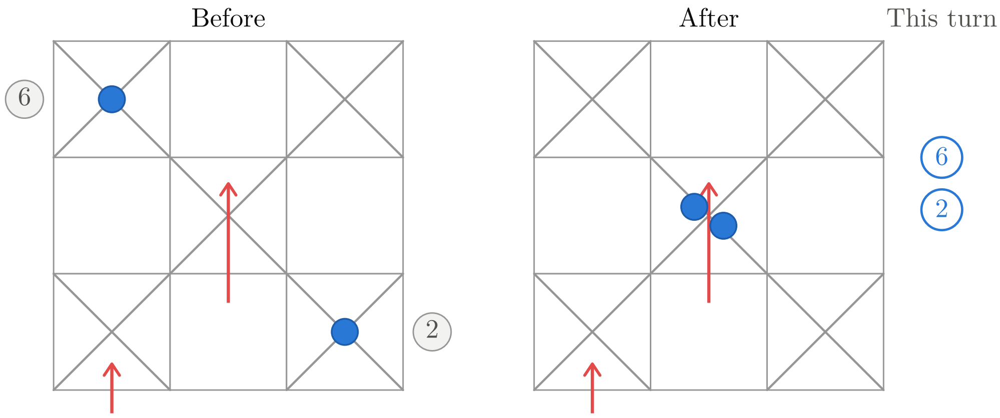
A throw of 6 or 12 always earns an immediate extra throw (Rule 2), even when that 6 or 12 itself overshoots and can’t bring a piece home. A later throw in that same chain — alone or summed with earlier ones — can still complete the piece’s journey if it lands exactly; if instead it falls short, it is not forced onto this piece and follows the same cases above.

An exact throw lands the piece on the centre



This turn produced 2 (not a 6 or 12, so no bonus throw follows). The piece is exactly 2 away, so the 2 lands it precisely on the centre.

Two pieces close enough: the player chooses which value goes where



A throw of 6 earns a bonus throw of 2, so this turn produced 6 and 2 — and both land a piece exactly on the centre. The player is free to assign either value to either piece.

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